

Literature status: strictly singular operators and c_0 -factorization on Schreier spaces

Run `fa_banach_001`; source arXiv:1907.10645

Source question

Beanland, Kania, and Laustsen study the closed ideal lattice of $\mathcal{B}(X)$ when X is Tsirelson's space or a finite-order Schreier space. In Section 5 of [1], for $X = X[\mathcal{S}_n]$, $n \geq 1$, they write $\overline{\mathcal{G}}_{c_0}(X)$ for the norm closure of the ideal of operators on X that factor through c_0 . They observe that

$$\overline{\mathcal{G}}_{c_0}(X) \subseteq \bigcap \{ \mathcal{I} : \mathcal{I} \text{ is a non-trivial spatial ideal of } \mathcal{B}(X) \},$$

and then ask, among other things, whether

$$\mathcal{S}(X) \subseteq \overline{\mathcal{G}}_{c_0}(X).$$

5. SOME OPEN QUESTIONS

Theorem 1.2(ii) and its proof raise some natural questions. To state them concisely, let $X = X[\mathcal{S}_n]$ for some $n \in \mathbb{N}$, and denote the closure of the ideal of operators on X which factor through c_0 by $\overline{\mathcal{G}}_{c_0}(X)$; in the notation of the proof of Theorem 1.2(ii), $\overline{\mathcal{G}}_{c_0}(X) = \overline{\langle Q \rangle}$, and the argument given in its last paragraph shows that

$$\overline{\mathcal{G}}_{c_0}(X) \subseteq \bigcap \{ \mathcal{I} : \mathcal{I} \text{ is a non-trivial spatial ideal of } \mathcal{B}(X) \}.$$

However, we do not know whether this inclusion is proper. We also do not know whether $\mathcal{S}(X) \subseteq \overline{\mathcal{G}}_{c_0}(X)$.

Another, somewhat less precise, question is as follows. It applies to both $X = T$ and $X = X[\mathcal{S}_n]$ for $n \in \mathbb{N}$. Theorem 1.1(iii) states that $\mathcal{B}(X)$ contains at least continuum

Figure 1: arXiv:1907.10645, page 22: the source question about $\mathcal{S}(X) \subseteq \overline{\mathcal{G}}_{c_0}(X)$ for $X = X[\mathcal{S}_n]$.

Supporting literature status

Manoussakis and Pelczar-Barwacz later studied small closed ideals on the Schlumprecht and Schreier spaces. In Section 4.2 of [2], they explicitly say that for finite-order Schreier spaces $X[\mathcal{S}_N]$, the reverse inclusion fails when $N \geq 2$, answering one of the questions of [1]; they also state that the case $N = 1$ remains open.

Proof intuition

The source question asks whether every strictly singular operator on a finite-order Schreier space lies in the closed ideal generated by operators factoring through c_0 . The later paper finds a canonical

Theorem 4.4. *Fix $N \in \mathbb{N}$, $N \geq 1$. There is a family of small closed operator ideals $(I_A)_{A \subset \mathbb{R}}$ on the Schreier space $X[\mathcal{S}_N]$ such that $I_A \subset I_B \Leftrightarrow A \subset B$, for any $A, B \subset \mathbb{R}$. In particular there are exactly $2^{\mathfrak{c}}$ distinct small closed ideals on the Schlumprecht space. There is also a chain of cardinality $\mathfrak{c}$ of small closed ideals and an antichain of cardinality $2^{\mathfrak{c}}$ of small closed ideals on the Schreier space $X[\mathcal{S}_N]$.*

Proof. Take a family $(T_{J_r})_{r \in \mathbb{R}} \subset \mathcal{S}(X_S)$ with $(J_r)_{r \in \mathbb{R}}$ as in Lemma 4.3 and for any $A \subset \mathbb{R}$ define I_A to be the closed operator ideal generated by $(T_{J_r})_{r \in A}$. Then obviously $A \subset B$ implies $I_A \subset I_B$. On the other hand, fix $A \subset \mathbb{R}$ and $r \in \mathbb{R} \setminus A$. Then by the choice of $(J_r)_{r \in \mathbb{R}}$ sets J_r and $(J_s)_{s \in A}$ satisfy the assumptions of Prop. 4.2 thus in particular $T_{J_r} \notin I_A$. In other words $T_{J_r} \in I_A$ implies $r \in A$, which finishes the proof. $\square$

4.2. Operators factoring through c_0 . As $X[\mathcal{S}_N]$ contains a copy of c_0 , by Sobczyk theorem the ideal $\overline{\mathcal{G}}_{c_0}(X[\mathcal{S}_N])$, i.e. the closure of the ideal of operators factoring through c_0 is not contained in $\mathcal{S}(X[\mathcal{S}_N])$. We show below that in the case $N \geq 2$ the reverse inclusion does not hold either, answering one of the questions of [BKL]. The case $N = 1$ remains open.

Figure 2: arXiv:2008.12362, page 14: the later paper states that the reverse inclusion fails for $N \geq 2$, answering one of the questions of [1], while $N = 1$ remains open.

obstruction: the formal identity $R_N : X[\mathcal{S}_N] \rightarrow X[\mathcal{S}_{N-1}]$. It is strictly singular, but for $N \geq 2$ it cannot be approximated within distance one by any composition through c_0 . Thus R_N is a strictly singular operator that remains outside $\overline{\mathcal{G}}_{c_0}(X[\mathcal{S}_N])$.

The literature answer

Theorem 1. *For every $N \in \mathbb{N}$ with $N \geq 2$,*

$$\mathcal{S}(X[\mathcal{S}_N]) \not\subset \overline{\mathcal{G}}_{c_0}(X[\mathcal{S}_N]).$$

Consequently, the inclusion question from [1] has a negative answer for finite-order Schreier spaces of order at least two.

Proof. Manoussakis and Pelczar-Barwacz prove in Proposition 4.6 of [2] that the formal identity

$$R_N : X[\mathcal{S}_N] \longrightarrow X[\mathcal{S}_{N-1}]$$

is strictly singular for every $N \geq 1$. They further prove that, if $N \geq 2$, then for every pair of bounded operators

$$T : X[\mathcal{S}_N] \rightarrow c_0, \quad S : c_0 \rightarrow X[\mathcal{S}_{N-1}],$$

one has

$$\|R_N - S \circ T\| \geq 1.$$

This says precisely that R_N cannot be approximated by operators that factor through c_0 . Since R_N is strictly singular, R_N gives an element of $\mathcal{S}(X[\mathcal{S}_N])$ that does not belong to $\overline{\mathcal{G}}_{c_0}(X[\mathcal{S}_N])$. Corollary 4.7 of [2] records the result explicitly:

$$\mathcal{S}(X[\mathcal{S}_N]) \not\subset \overline{\mathcal{G}}_{c_0}(X[\mathcal{S}_N]) \quad (N \geq 2).$$

$$\bigcup_{n > n_0} Z''$$

As $K \in \mathbb{N}$ is arbitrary this yields contradiction and ends the proof of strict singularity of R_N .

Now let $N \geq 2$. Assume for some bounded operators $T : X[\mathcal{S}_N] \rightarrow c_0$ and $S : c_0 \rightarrow X[\mathcal{S}_{N-1}]$ we have $\|R_N - S \circ T\| < 1$. As $R_N(\tilde{e}_n)(n) = 1$, $n \in \mathbb{N}$, it follows that

$$\delta := \inf_n (S \circ T)\tilde{e}_n(n) > 0$$

As $(\tilde{e}_n)_n$ is weakly null, $(T\tilde{e}_n)_n$ and $((S \circ T)\tilde{e}_n)_n$ are also weakly null, thus we can choose some infinite $M \subset \mathbb{N}$ such that the sequence $(T\tilde{e}_n)_{n \in M}$ is equivalent to the unit vector basis of c_0 , whereas $|((S \circ T)\tilde{e}_n)(m)| < \frac{1}{2^{n+m+1}}$ for any $n, m \in M$, $n \neq m$.

Let $P_M : X[\mathcal{S}_{N-1}] \rightarrow [\tilde{e}_n : n \in M] \subset X[\mathcal{S}_{N-1}]$ be the canonical basis projection. By the choice of M we have $(P_M \circ S)(T\tilde{e}_n) = \alpha_n \tilde{e}_n + w_n$ with $\alpha_n \geq \delta$ and $\|w_n\|_{\mathcal{S}_{N-1}} \leq \|w_n\|_{\ell_1} \leq \frac{1}{2^n}$ for any $n \in M$. Therefore the sequence $(\tilde{e}_n)_{n \in M}$ in $X[\mathcal{S}_{N-1}]$, as equivalent to $((P_M \circ S)(T\tilde{e}_n))_{n \in M}$, is dominated by $(T\tilde{e}_n)_{n \in M}$. Thus we obtain a contradiction, as $(T\tilde{e}_n)_{n \in M}$ is equivalent to the unit vector basis of c_0 . $\square$

Corollary 4.7. $\mathcal{S}(X[\mathcal{S}_N]) \not\subseteq \overline{\mathcal{G}}_{c_0}(X[\mathcal{S}_N])$ for any $N \in \mathbb{N}$, $N \geq 2$.

REFERENCES

Figure 3: arXiv:2008.12362, page 16: Corollary 4.7 gives $\mathcal{S}(X[\mathcal{S}_N]) \not\subseteq \overline{\mathcal{G}}_{c_0}(X[\mathcal{S}_N])$ for $N \geq 2$. $\square$

Verification notes and limitations

This packet is a literature-status identification, not a new proof from the run. The supporting paper explicitly identifies the result as answering one of the questions of [1]. The evidence crops above show both the source question and the later answer/corollary.

The result is a partial subcase of the Section 5 questions in [1]. It does not settle $N = 1$; [2] explicitly leaves that case open. It also does not decide whether $\overline{\mathcal{G}}_{c_0}(X)$ is properly contained in the intersection of all non-trivial spatial ideals, nor does it describe explicit maximal ideals in $\mathcal{B}(T)$ or $\mathcal{B}(X[\mathcal{S}_n])$.

Before promotion, the run lightweight indexes were searched for arXiv:1907.10645, the title phrase, “closed ideals of operators”, “Tsirelson”, “Schreier”, “operator ideals”, and nearby core terms. No existing exact packet or attempt was found. Local source search and web search then identified arXiv:2008.12362 as the relevant later paper.

Human review recommendation

Treat the source question $\mathcal{S}(X) \subseteq \overline{\mathcal{G}}_{c_0}(X)$ as already answered negatively for $X = X[\mathcal{S}_N]$, $N \geq 2$, by [2]. Keep the $N = 1$ case and the other open questions from the source Section 5 separate.

References

- [1] K. Beanland, T. Kania, and N. J. Laustsen, *Closed ideals of operators on the Tsirelson and Schreier spaces*, J. Funct. Anal. 279 (2020), 108668; arXiv:1907.10645.
- [2] A. Manoussakis and A. Pelczar-Barwacz, *Small operator ideals on the Schlumprecht and Schreier spaces*, arXiv:2008.12362.